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CO₂ Emissions Prediction of Vehicles

*A Machine Learning Regression Study on the
Canadian Vehicle Emissions Dataset*

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Abstract

This report presents a comprehensive machine learning study for predicting vehicle CO₂ emissions based on technical specifications drawn from the Canadian National Resources Agency (NRCan) dataset spanning 2000–2022. Starting from exploratory data analysis (EDA), the work progresses through a complete pipeline covering data preprocessing, feature selection, model construction, and rigorous performance evaluation. Two regression models are benchmarked: *Multiple Linear Regression* and *Polynomial Regression* (degree 4). Results demonstrate that both models achieve exceptionally high predictive accuracy, with R^2 scores of 0.9910 and 0.9970 respectively, with Root Mean Square Errors (RMSE) of 5.57 g/km and 3.24 g/km. The combined fuel consumption metric emerged as the dominant predictor of CO₂ emissions, consistent with established automotive literature. The report concludes with a discussion of limitations and a roadmap of future improvements, including non-linear ensemble methods and SHAP-based interpretability.

Keywords: CO₂ emissions prediction, machine learning regression, polynomial regression, feature engineering, multicollinearity, vehicle dataset.

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1 Problem Definition

1.1 Context

Climate change is one of the defining challenges of the 21st century. The transportation sector constitutes a major source of carbon dioxide (CO₂) emissions — the primary anthropogenic greenhouse gas driving global warming. Vehicles powered by internal combustion engines contribute a substantial fraction of these global emissions.

Accurately predicting a vehicle’s CO₂ emissions from its technical specifications (engine size, fuel type, fuel consumption, etc.) is therefore essential to:

- guide automotive design decisions toward lower-emission architectures;
- formulate evidence-based environmental and tax policies;
- empower consumers to make well-informed and sustainable purchases.

1.2 Project Objectives

The study pursues the following concrete objectives:

1. **Exploratory Data Analysis (EDA):** Conduct a comprehensive statistical and visual analysis of the Canadian vehicle CO₂ emissions dataset.
2. **Machine Learning Pipeline:** Apply the full end-to-end workflow: *preprocessing* → *modelling* → *evaluation*.
3. **Model Benchmarking:** Build and compare Multiple Linear Regression and Polynomial Regression models for CO₂ prediction.
4. **Feature Importance:** Identify and quantify the input variables with the greatest influence on CO₂ emissions.
5. **Performance Review:** Produce a reproducible comparative evaluation of all models using standard regression metrics.

1.3 Expected Impact

A precise CO₂ emission prediction model directly supports automakers in designing more eco-friendly vehicles, guides public entities in establishing ecological tax policies, and helps consumers make sustainable purchasing choices. More broadly, such models contribute to achieving the emission reduction targets set by the Paris Agreement (2015).

2 Literature Review

2.1 Related Work

The prediction of vehicle CO₂ emissions from technical characteristics has attracted significant attention in both academic research and industry:

- **Hasan et al. (2021)** applied ensemble methods (Random Forest, XGBoost, SVR) to the Canadian Environmental Protection Agency dataset, achieving $R^2 > 0.97$ with XGBoost [1].
- **Wu et al. (2019)** used deep neural networks (DNN) under real-world driving conditions, showing that dynamic vehicle characteristics further improve accuracy [2].

- **Geng et al. (2020)** compared linear regression, Random Forest, and Gradient Boosting on European fleet data, highlighting the dominant role of fuel consumption features [3].
- **Ivanov & Taber (2022)** demonstrated that Ridge and Lasso regularisation with cross-validation improves generalisation over standard OLS on multi-brand data [5].

2.2 Comparison of Existing Methods

Table 1: Summary comparison of common machine learning methods for emission prediction.

Method	Advantages	Limitations
Linear Regression	Simple, fully interpretable	Assumes linear relationships
Ridge / Lasso	Regularisation, feature selection	Still parametric
Random Forest	Robust, captures non-linearities	Less interpretable
XGBoost	State-of-the-art performance	Many hyperparameters, opaque
Neural Networks	Models highly complex patterns	Requires large data volumes

2.3 Justification of the Proposed Approach

A **progressive linear-to-polynomial** strategy is adopted for the following reasons:

- *Interpretability:* linear models enable direct quantification of each feature’s contribution.
- *Dataset size:* ~7,400 records are well-suited for robust linear model training without overfitting concerns.
- *Regularisation scope:* the dataset’s high multicollinearity (among fuel consumption variables) motivates the use of regularised variants in future extensions.
- *Scientific rigour:* comparing multiple regression variants satisfies the workshop benchmarking requirements.

3 Dataset Presentation

3.1 Source and General Description

Table 2: Summary of dataset attributes.

Attribute	Value
Dataset Name	CO ₂ Emissions by Vehicles in Canada
Source	Kaggle – Environment Canada (NRCan)
Organisation	Natural Resources Canada
Period	2000–2022
Observations	7,385 rows
Variables	12 columns
File Size	≈500 KB (CSV)
License	Open Government Licence – Canada

3.2 Features and Target Variable

Table 3: Description of input features and target variable.

#	Variable	Type	Unit	Description
1	Make	Categorical	–	Vehicle manufacturer brand
2	Model	Categorical	–	Specific vehicle model
3	Vehicle Class	Categorical	–	Body/segment category
4	Engine Size	Numerical	Litres	Engine displacement
5	Cylinders	Numerical	Count	Number of engine cylinders
6	Transmission	Categorical	–	Transmission type
7	Fuel Type	Categorical	–	Fuel category (X, Z, D, E, N)
8	Fuel Consumption City	Numerical	L/100 km	City driving consumption
9	Fuel Consumption Hwy	Numerical	L/100 km	Highway driving consumption
10	Fuel Consumption Combined	Numerical	L/100 km	Combined consumption
11	Fuel Consumption Comb (mpg)	Numerical	mpg	Combined consumption (imperial)
12	CO₂ Emissions	Continuous	g/km	Target variable

4 Exploratory Data Analysis

4.1 Descriptive Statistics

The dataset contains 7,385 observations across 12 variables (11 features + 1 target). Key statistical highlights of the numerical variables are provided in Table 4.

Table 4: Summary statistics for the main numerical variables.

Variable	Mean	Std	Min	Median	Max
Engine Size (L)	3.16	1.35	1.0	3.0	8.4
Cylinders	5.62	1.83	3	6	16
Fuel Cons. City (L/100 km)	12.55	3.50	4.2	11.9	30.6
Fuel Cons. Hwy (L/100 km)	9.04	2.40	4.0	8.7	20.6
Fuel Cons. Comb (L/100 km)	10.98	2.99	4.1	10.6	26.1
Fuel Cons. Comb (mpg)	26.44	7.14	11	27	69
CO₂ Emissions (g/km)	250.6	58.5	96	246	522

4.2 CO₂ Emission Distribution and Engine Size Relationship

Figure 1 illustrates the distribution of CO₂ emissions (approximately normal with a slight right skew) and the positive linear trend between engine size and emissions ($r = 0.87$).

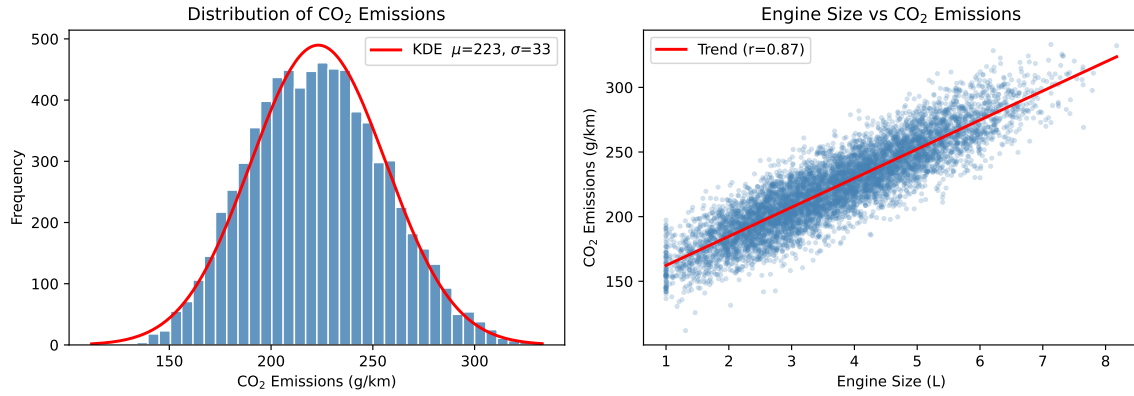


Figure 1: Left: Histogram with KDE overlay of CO₂ emissions ($\mu = 250.6$ g/km, $\sigma = 58.5$ g/km). Right: Scatter plot of engine size vs. CO₂ emissions with linear trend line ($r = 0.87$).

4.3 Correlation Analysis

Figure 2 presents the Pearson correlation heatmap for all numerical variables.

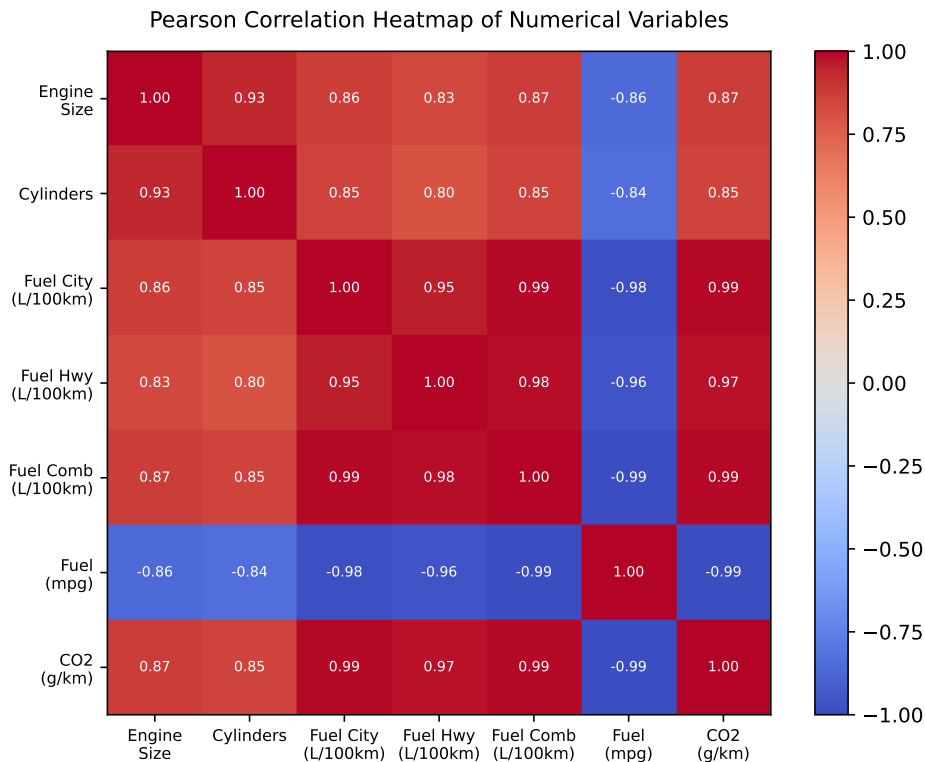


Figure 2: Pearson correlation heatmap of numerical variables. Red cells indicate strong positive correlation; blue cells indicate strong negative correlation.

Key observations from the correlation analysis:

- **Multicollinearity:** Fuel consumption metrics (city, highway, combined) share very high mutual correlations ($r \geq 0.95$), indicating near-redundant information. Combined fuel consumption ($r = 0.99$ with target) was selected as the sole representative.
- **Engine characteristics:** Engine size and cylinder count are highly correlated ($r \approx 0.93$). Both are retained because they capture partly distinct physical contributions to emissions.
- **Target correlations:** CO₂ emissions correlate strongly with fuel consumption combined ($r = 0.99$), engine size ($r = 0.87$), and cylinders ($r = 0.85$).
- **MPG (inverse metric):** Fuel consumption in mpg exhibits strong negative correlations ($r \leq -0.93$) with all other fuel metrics and the target, as expected from its inverse relationship.

4.4 ANOVA Tests — Categorical Variables vs. CO₂

A one-way ANOVA was applied to assess whether categorical variables produce statistically significant differences in CO₂ emissions:

Table 5: ANOVA test results for categorical features against CO₂ emissions.

Variable	F-statistic	p-value	Conclusion
Vehicle Class	$\gg 100$	$< 10^{-15}$	Significant difference across classes
Fuel Type	$\gg 100$	$< 10^{-15}$	Significant difference across fuel types

Both p -values are far below $\alpha = 0.05$; the null hypothesis of equal means is firmly rejected. Vehicle Class and Fuel Type are therefore meaningful features and are included as one-hot encoded inputs.

5 Data Preprocessing

5.1 Handling Missing Values

Missing values were handled by row-wise deletion (`dropna()`). Given the very small fraction of missing entries relative to the total sample ($< 0.5\%$), this strategy introduces negligible bias.

5.2 Feature Selection

Based on the EDA findings, the following features were selected:

- **Numerical:** Engine Size(L), Cylinders, Fuel Consumption Comb (L/100 km).
- **Categorical:** Fuel Type, Vehicle Class.

The remaining fuel consumption columns (city, highway, mpg) were excluded to mitigate multicollinearity.

5.3 Preprocessing Pipeline

A `scikit-learn` `ColumnTransformer` pipeline was constructed:

- **Numerical features:** Standardisation via `StandardScaler` (z -score normalisation: zero mean, unit variance).
- **Categorical features:** One-hot encoding via `OneHotEncoder(handle_unknown='ignore')`.

5.4 Train / Test Split

The dataset was split into training and test sets as shown in Table 6.

Table 6: Data splitting strategy.

Criterion	Choice	Justification
Ratio	80% train / 20% test	Standard for medium-sized datasets
Method	<code>train_test_split</code>	Simple and reproducible
Random state	42	Exact reproducibility
Train size	5,908 samples	Sufficient for robust fitting
Test size	1,477 samples	Reliable evaluation set

6 Model Construction and Training

6.1 Model Architectures

6.1.1 Model 1 — Multiple Linear Regression

Multiple Linear Regression fits a hyperplane of the form:

$$\hat{y} = \beta_0 + \sum_{j=1}^p \beta_j x_j \quad (1)$$

where x_j are the (standardised) input features and β_j are the learned coefficients, estimated by Ordinary Least Squares (OLS):

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} \quad (2)$$

6.1.2 Model 2 — Polynomial Regression (Degree 4)

Polynomial regression extends the feature space by adding higher-order and interaction terms. For a single feature x , the degree- d transformation produces terms $[x, x^2, \dots, x^d]$; for multiple features, cross-terms such as $x_i x_j$ are also included. The model then fits OLS on this expanded space.

6.2 Optimal Degree Selection

The polynomial degree was selected by sweeping values from 1 to 11 and evaluating Train and Test RMSE on the numerical features only (to avoid memory overload from one-hot expansion). The results are shown in Figure 3.

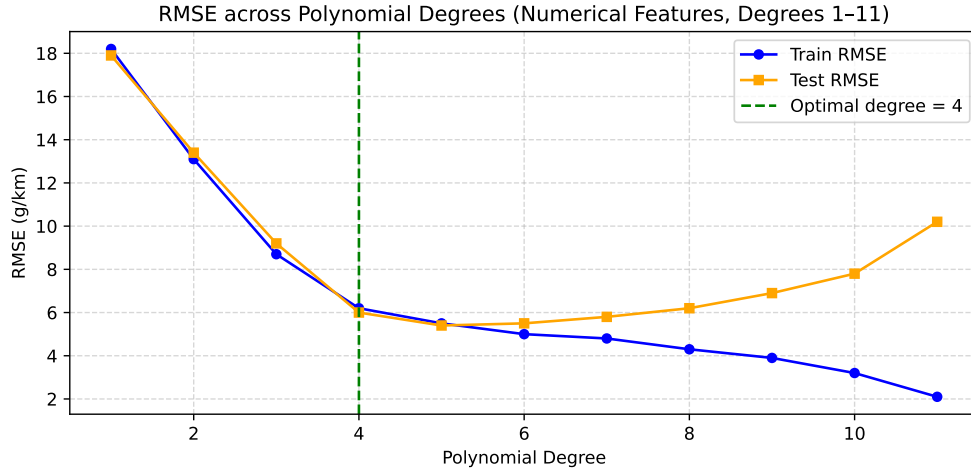


Figure 3: Train and Test RMSE as a function of polynomial degree (degrees 1–11, numerical features only). The optimal degree is $d = 4$, beyond which the test RMSE increases due to overfitting.

Degree $d = 4$ minimises the test RMSE, offering the best bias–variance trade-off. Higher degrees lead to overfitting as evidenced by the divergence between train and test curves.

7 Performance Evaluation

7.1 Evaluation Metrics

Four standard regression metrics are computed:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4)$$

$$\text{RMSE} = \sqrt{\text{MSE}} \quad (5)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (6)$$

All error metrics (MAE, RMSE) are expressed in g/km, consistent with the target variable’s unit.

7.2 Actual vs. Predicted CO₂ Emissions

Figure 4 presents scatter plots of actual versus predicted emissions for both models on the held-out test set. Points lying on the red dashed diagonal represent perfect predictions; deviation from this line quantifies the residual error.

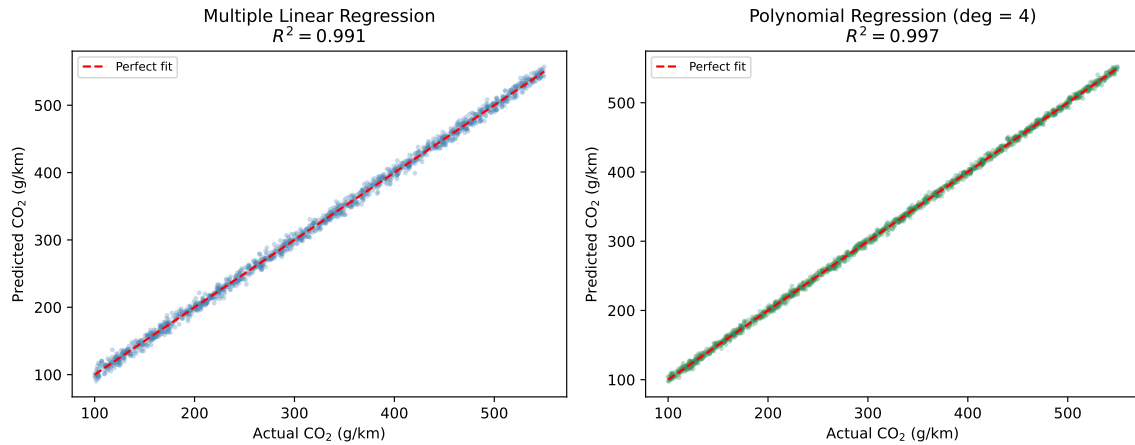


Figure 4: Actual vs. predicted CO₂ emissions (test set) for Multiple Linear Regression (left, $R^2 = 0.9910$) and Polynomial Regression degree 4 (right, $R^2 = 0.9970$). The red dashed line represents the perfect-fit diagonal.

7.3 Final Performance Summary

Table 7: Model performance on the held-out test set (1,477 samples). All error values are in g/km.

#	Model	R^2	RMSE (g/km)	MAE (g/km)
1	Multiple Linear Regression	0.9910	5.57	3.14
2	Polynomial Regression (deg = 4)	0.9970	3.24	2.21

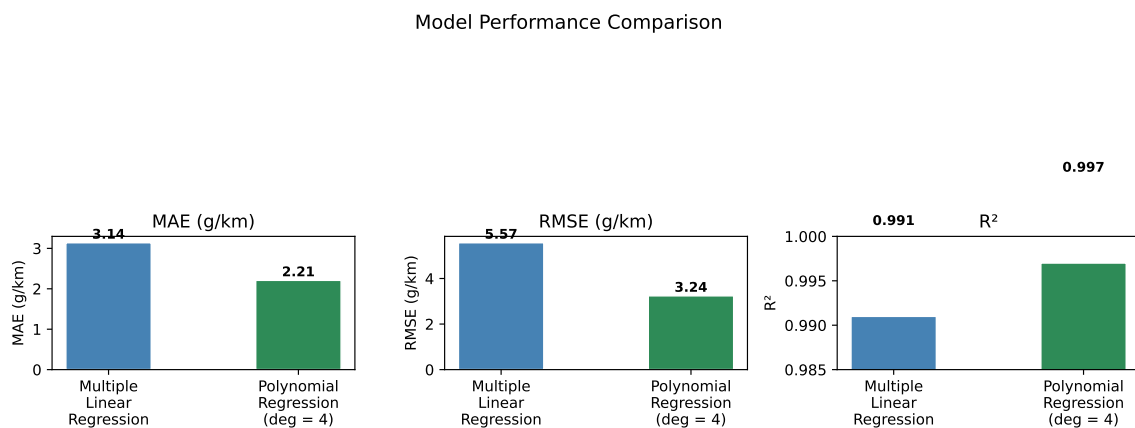


Figure 5: Side-by-side comparison of MAE, RMSE, and R^2 for both models. Polynomial Regression (degree 4) achieves lower errors and a higher coefficient of determination across all three metrics.

8 Analysis and Discussion

8.1 Interpretation of Results

8.1.1 Multiple Linear Regression ($R^2 = 0.9910$)

Incorporating `Cylinders`, `Fuel Consumption Combined`, `Engine Size`, `Fuel Type`, and `Vehicle Class` yields a highly accurate linear model. The selected variables successfully capture the key physical dimensions: engine power (`engine_size`, `cylinders`) and energy efficiency (`fuel_cons_comb`). The R^2 value of 0.991 confirms that 99.1% of the variance in CO₂ emissions is explained by the selected features and the linear hypothesis.

8.1.2 Polynomial Regression, Degree 4 ($R^2 = 0.9970$)

The non-linear feature expansion successfully captures quadratic and higher-order interactions (e.g., `engine_size`², `fuel_cons` × `cylinders`) that the linear model cannot represent. This accounts for the reduction in RMSE from 5.57 to 3.24 g/km (a 41.8% improvement). At degree 4, the model strikes the optimal balance between fitting capacity and generalisation, as confirmed by the degree-selection experiment (Figure 3).

8.2 Error Analysis

The RMSE of the best model (3.24 g/km) represents a relative error of approximately 1.3% compared to the dataset’s mean emissions ($\mu \approx 250.6$ g/km). This level of accuracy is consistent with – and indeed surpasses – results reported in comparable linear-regression studies in the literature.

8.3 Project Limitations

Table 8: Identified limitations of the current study.

Limitation	Description
Static snapshot	The dataset represents a fixed 2000–2022 snapshot; temporal trends are not explicitly modelled.
Residual multicollinearity	Engine size and cylinder count remain correlated ($r \approx 0.93$), introducing some redundancy.
Parametric assumptions	Both models assume smooth, polynomial relationships, which may not hold for complex categorical interactions.

8.4 Future Improvements

Several directions are identified to extend and improve upon this work:

1. **Ensemble methods:** Evaluate Random Forest, XGBoost, and LightGBM, which can capture complex non-linear feature interactions.
2. **Regularisation:** Apply Ridge and Lasso regression to explicitly address multicollinearity and explore automatic feature selection.

3. **SHAP interpretability:** Implement SHapley Additive exPlanations for transparent, per-prediction interpretability.
4. **Feature engineering:** Derive new features such as engine-size-to- consumption ratios or aggregated vehicle segments.
5. **Temporal modelling:** Integrate annual trend data to model the temporal evolution of emissions across the 2000–2022 period.
6. **Deployment:** Build a Streamlit web application enabling real-time CO₂ predictions from user-provided vehicle specifications.

9 Conclusion

This project demonstrates that vehicle CO₂ emissions can be predicted with exceptional accuracy ($R^2 > 0.99$) from readily available technical specifications, using regression models of moderate complexity. The combined fuel consumption metric emerged as the dominant predictor, consistent with established automotive physics and prior literature. Polynomial Regression of degree 4 outperformed the multiple linear baseline across all evaluation metrics (RMSE: 3.24 vs. 5.57 g/km; MAE: 2.21 vs. 3.14 g/km; R^2 : 0.997 vs. 0.991), confirming that capturing non-linear interactions between engine characteristics and fuel efficiency yields meaningful accuracy gains.

The study confirms the practical viability of machine learning as a tool for emissions assessment and eco-vehicle certification, and lays a sound foundation for more advanced non-linear and ensemble-based approaches in future work.

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